

5. A light elastic string has natural length $2a$ and modulus of elasticity $2mg$. One end of the string is attached to a fixed point A on a horizontal ceiling. The other end is attached to a particle P of mass m .

The particle P hangs in equilibrium at the point E , where $AE = 3a$.

The particle P is then projected vertically downwards from E with speed $\frac{3}{2}\sqrt{ag}$

Air resistance is assumed to be negligible.

Find the elastic energy stored in the string, when P first comes to instantaneous rest. Give your answer in the form $kmg a$, where k is a constant to be found.

(7)

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4. A light elastic string has natural length $2a$ and modulus of elasticity $4mg$.
One end of the elastic string is attached to a fixed point O . A particle P of mass m is attached to the other end of the elastic string.
The particle P hangs freely in equilibrium at the point E , which is vertically below O

(a) Find the length OE .

(4)

Particle P is now pulled vertically downwards to the point A , where $OA = 4a$, and released from rest. The resistance to the motion of P is a constant force of magnitude $\frac{1}{4}mg$.

(b) Find, in terms of a and g , the speed of P after it has moved a distance a .

(7)

Particle P is now held at O

Particle P is released from rest and reaches its maximum speed at the point B .

The resistance to the motion of P is again a constant force of magnitude $\frac{1}{4}mg$.

(c) Find the distance OB .

(4)

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6. A light elastic string with natural length l and modulus of elasticity kmg has one end attached to a fixed point A on a rough inclined plane. The other end of the string is attached to a package of mass m .

The plane is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{5}{12}$

The package is initially held at A . The package is then projected with speed $\sqrt{6gl}$ up a line of greatest slope of the plane and first comes to rest at the point B , where $AB = 3l$.

The coefficient of friction between the package and the plane is $\frac{1}{4}$

By modelling the package as a particle,

- (a) show that $k = \frac{15}{26}$ (6)
- (b) find the acceleration of the package at the instant it starts to move back down the plane from the point B . (5)

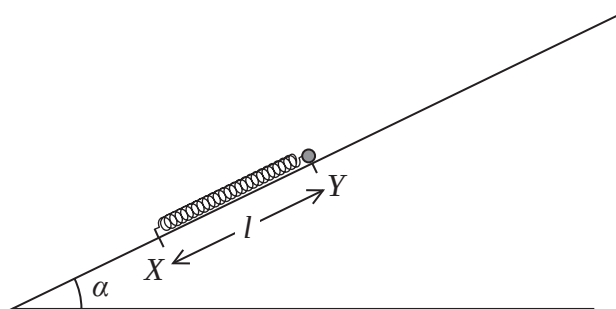
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6.

**Figure 2**

A light elastic spring has natural length $3l$ and modulus of elasticity $3mg$.

One end of the spring is attached to a fixed point X on a rough inclined plane.

The other end of the spring is attached to a package P of mass m .

The plane is inclined to the horizontal at an angle α where $\tan \alpha = \frac{3}{4}$

The package is initially held at the point Y on the plane, where $XY = l$. The point Y is higher than X and XY is a line of greatest slope of the plane, as shown in Figure 2.

The package is released from rest at Y and moves up the plane.

The coefficient of friction between P and the plane is $\frac{1}{3}$

By modelling P as a particle,

(a) show that the acceleration of P at the instant when P is released from rest is $\frac{17}{15}g$ (5)

(b) find, in terms of g and l , the speed of P at the instant when the spring first reaches its natural length of $3l$. (6)

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7. A particle P , of mass m , is attached to one end of a light elastic spring of natural length a and modulus of elasticity kmg .

The other end of the spring is attached to a fixed point O on a ceiling.

The point A is vertically below O such that $OA = 3a$

The point B is vertically below O such that $OB = \frac{1}{2}a$

The particle is held at rest at A , then released and first comes to instantaneous rest at the point B .

- (a) Show that $k = \frac{4}{3}$ (3)

- (b) Find, in terms of g , the acceleration of P immediately after it is released from rest at A . (3)

- (c) Find, in terms of g and a , the maximum speed attained by P as it moves from A to B . (6)

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7. A particle P of mass m is attached to one end of a light elastic string of natural length a and modulus of elasticity $3mg$.

The other end of the string is attached to a fixed point O on a ceiling.

The particle hangs freely in equilibrium at a distance d vertically below O .

- (a) Show that $d = \frac{4}{3}a$. (3)

The point A is vertically below O such that $OA = 2a$.

The particle is held at rest at A , then released and first comes to instantaneous rest at the point B .

- (b) Find, in terms of g , the acceleration of P immediately after it is released from rest. (3)

- (c) Find, in terms of g and a , the maximum speed attained by P as it moves from A to B . (5)

- (d) Find, in terms of a , the distance OB . (3)

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